

Overview of GPT language model

Subject: GPT language model

1.

Efficient transformer architectures The computational and memory requirements of transformer-based models increase significantly as they scale to larger sizes and longer input sequences. The standard self-attention mechanism has a quadratic complexity with respect to input sequence length, as described in Attention Is All You Need. Researchers proposed a number of efficiency improvements like sparse attention mechanisms and memory-efficient architectures that reduce computational costs while supporting longer context windows. Models like BigBird, Reformer, and FlashAttention demonstrate structured attention patterns or optimized computation to improve scalability and efficiency. This has helped large language models to efficiently process long input sequences with reduced memory and computation during both training and inference.

2.

Background During the 2010s, improved machine learning algorithms, more powerful computers, and an increase in the amount of digitized material allowed for an AI boom. Separately, the concept of generative pre-training (GP) was a long-established technique in machine learning. GP is a form of self-supervised learning wherein a model is first trained on a large, unlabeled dataset (the "pre-training" step) to learn to generate data points. This pre-trained model is then adapted to a specific task using a labeled dataset (the "fine-tuning" step). The transformer architecture for deep learning is the core technology of a GPT. Developed by researchers at Google, it was introduced in the paper "Attention Is All You Need", which was released on June 12, 2017. The transformer architecture solved many of the performance issues that were associated with older recurrent neural network (RNN) designs for natural language processing (NLP). The architecture's use of an attention mechanism allows models to process entire sequences of text at once, enabling the training of much larger and more sophisticated models. Since 2017, available transformer-based NLP systems have been capable of processing, mining, organizing, connecting, contrasting, and summarizing texts as well as answering questions from textual input.

3.

Foundation models A foundation model is an AI model trained on broad data at scale such that it can be adapted to a wide range of downstream tasks. The most recent OpenAI's GPT-n series model is GPT-5. Other such models include Google's PaLM, a broad foundation model that has been compared to GPT-3 and has been made available to developers via an API, and Together's GPT-JT, which has been reported as the closest-performing open-source alternative to GPT-3 (and is derived from earlier open-source GPTs). Meta AI (formerly Facebook) also has a generative transformer-based foundational large language model, known as LLaMA. Foundational GPTs can also employ modalities other than text, for input and/or output. GPT-4 is a multi-modal LLM that is capable of processing text and image input (though its output is limited to text). Regarding multimodal output, some generative transformer-based models are used for text-to-image technologies such as diffusion and parallel decoding. Such kinds of models can serve as visual foundation models (VFMs) for developing downstream systems that can work with images.

4.

A generative pre-trained transformer (GPT) is a type of large language model (LLM) that is widely used in generative artificial intelligence chatbots. GPTs are based on a deep learning architecture called the transformer. They are pre-trained on large datasets of unlabeled content, and able to generate novel content. OpenAI was the first to apply generative pre-training to the transformer architecture, introducing the GPT-1 model in 2018. The company has since released many bigger GPT models. The chatbot ChatGPT, released in late 2022 (using GPT-3.5), was followed by many competitor chatbots using their own generative pre-trained transformers to generate text, such as Gemini, DeepSeek and Claude. GPTs are primarily used to generate text, but can be trained to generate other kinds of data. For example, GPT-4o can process and generate text, images and audio. To improve performance on complex tasks, some GPTs, such as OpenAI o3, allocate more computation time analyzing the problem before generating an output, and are called reasoning models. In 2025, GPT-5 was released with a router that automatically selects whether to use a faster model or slower reasoning model based on the provided task.

5.

OpenAI, which created the first generative pre-trained transformer (GPT) in 2018, asserted in 2023 that "GPT" should be regarded as a brand of OpenAI. In April 2023, OpenAI revised the brand guidelines in its terms of service to indicate that other businesses using its API to run their AI services would no longer be able to include "GPT" in such names or branding. In May 2023, OpenAI engaged a brand management service to notify its API customers of this policy, although these notifications stopped short of making overt legal claims (such as allegations of trademark infringement or demands to cease and desist). As of November 2023, OpenAI still prohibits its API licensees from naming their own products with "GPT", but it has begun enabling its ChatGPT Plus subscribers to make "custom versions of ChatGPT" called GPTs on the OpenAI site. OpenAI's terms of service says that its subscribers may use "GPT" in the names of these, although it's "discouraged". Relatedly, OpenAI has applied to the United States Patent and Trademark Office (USPTO) to seek domestic trademark registration for the term "GPT" in the field of AI. OpenAI sought to expedite handling of its application, but the USPTO declined that request in April 2023. In May 2023, the USPTO responded to the application with a determination that "GPT" was both descriptive and generic. As of November 2023, OpenAI continued to pursue its argument. For any given type or scope of trademark protection in the U.S., OpenAI would need to establish that the term is actually "distinctive" to their specific offerings in addition to being a broader technical term for the kind of technology. Some media reports suggested in 2023 that OpenAI may be able to obtain trademark registration based indirectly on the fame of its GPT-based chatbot product, ChatGPT, for which OpenAI has separately sought protection (and which it has sought to enforce more strongly). Other reports have indicated that registration for the bare term "GPT" seems unlikely to be granted, as it is used frequently as a common term to refer simply to AI systems that involve generative pre-trained transformers. In any event, to whatever extent exclusive rights in the term may occur in the U.S., others would need to avoid using it for similar products or services in ways likely to cause confusion. If such rights ever became broad enough to implicate other well-established uses in the field, the trademark doctrine of descriptive fair use could still continue non-brand-related usage. In the European Union, the European Union Intellectual Property Office registered "GPT" as a trade mark of OpenAI in spring 2023. However, since spring 2024 the registration is being challenged and is pending cancellation. In Switzerland, the Swiss Federal Institute of Intellectual Property registered "GPT" as a trade mark of OpenAI in spring 2023.